Temporal Causal Knowledge Extraction, Transfer, and Discovery Prediction from Scientific Literature

Phase 1 Research Review: Multi-Page Extraction & Calibrated Grounding

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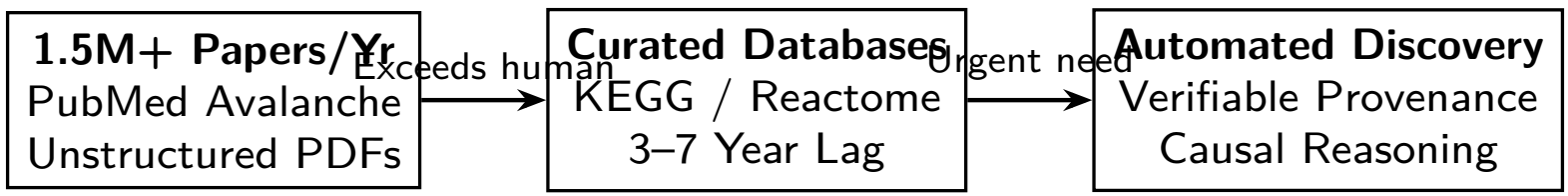
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Research Context: Automated Science & Biomedical Literature

Takeaway: Exponential literature growth overwhelms manual curation, requiring automated extraction.

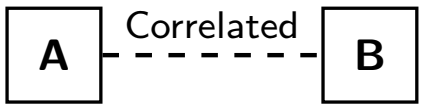


- **Curation Bottleneck**: Manual database synthesis captures $< 5\%$ of published biomedical mechanisms.
- **High-Stakes Domain**: Biomedical discovery requires verifiable, evidence-grounded facts, not hallucinations.

Motivation: Beyond Correlation to Grounded Causality

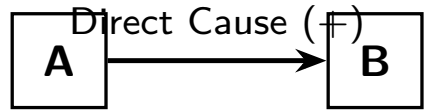
Takeaway: Therapeutic intervention requires causal directionality ($X \rightarrow Y$), which correlation cannot provide.

Associational Graph: $P(Y \mid X)$



- Symmetric co-occurrence
- Confounded by latent factors
- Fails Pearl's *do*-calculus

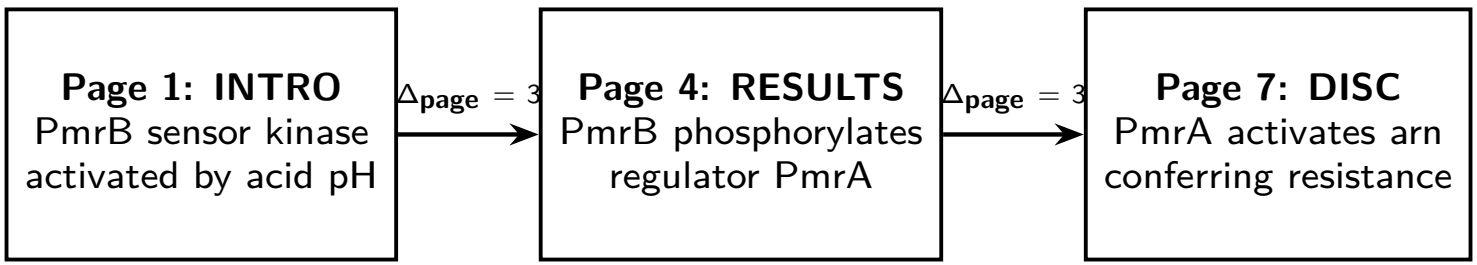
Structural Causal Graph: $P(Y \mid do(X))$



- Explicit direction and polarity
- Actionable target validation
- Verifiable textual provenance

Problem Statement: Multi-Page Causal Extraction

Takeaway: Scientific papers articulate mechanisms across physical pages and distant sections.



- **Physical Dispersion**: Sentence-local extractors miss 100% of cross-page mechanisms ($F_1^{\text{cross}} = 0.000$).
- **Semantic Drift**: Unconstrained cross-page search produces combinatorial false positive jumps across hub entities.

Research Gap: Limitations of Existing Approaches

Takeaway: Neither local NLP tools nor frontier LLMs resolve evidence-grounded multi-page causality.

Approach	Core Failure Mode	Observed Metric
Sentence NLP	Blind to cross-page discourse	$F_1^{\text{cross}} = 0.0000$
Statistical NPMI	Global co-occurrence; zero grounding	Grounding: 0.0%
Dense Embeddings	Semantic drift; no causal polarity	Grounding: 14.3%
Frontier LLMs	Hallucinates ungrounded causal links	ReCITE $F_1 = 0.535 \pm 0.20$

- **The Core Need**: An architecture that performs bounded multi-page extraction with 100% provenance traceability.

Research Objectives of Phase 1

Takeaway: Four formal engineering and research objectives established for Phase 1.

O1: Ingestion & Provenance
100% character offset tracing

O3: Hub Control & Invariants
Prune false transitive jumps

O2: Multi-Page Extraction
Direct and cross-page dependencies

O4: Strict Benchmark Auditing
20 papers across 4 isolated splits

- **Target**: Satisfy formal acceptance criteria with zero data leakage.

Research Questions Addressed in Phase 1

Takeaway: Investigating three foundational research questions in empirical literature extraction.

RQ1 (Grounding): Can deterministic syntactic parsing guarantee 100% evidence traceability without ha

RQ2 (Transitivity): How can transitive causal inference cross pages without triggering combinatorial false

RQ3 (Generalization): Does a bounded document-context model generalize to unseen biomedical lite

- All questions addressed through controlled experiments on 20 authentic PMC papers.

Scope and Contributions of Phase 1

Takeaway: Phase 1 delivers a verified modular pipeline, benchmark suite, and reliability engine.

Methodological Contributions

- Bounded 2-hop cross-page transitive search with distance-based confidence decay.
- Biological direction invariants (Terminal Phenotype & Environmental Input).
- Document-context relation resolver for demonstrative anaphora and split stitching.

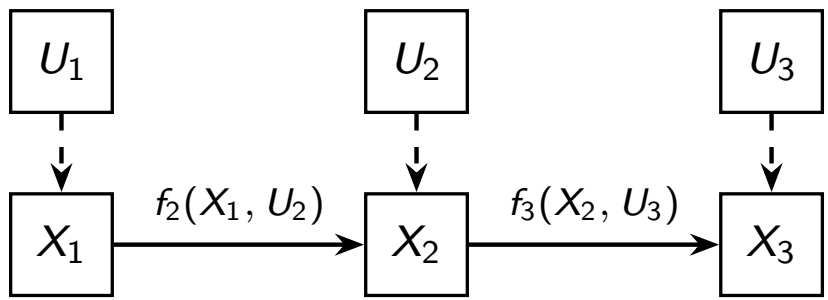
Engineering Artifacts

- 20-paper multi-split authentic PMC benchmark (V1 & V2 ground truth).
- 19-test automated reliability engine (`test_paper_reliability.py`).
- 100% reproducible evaluation suite with bootstrap confidence intervals.

Conceptual Foundations: Structural Causal Models (SCMs)

Takeaway: Pearl's 4-tuple formalizes the mathematical target of scientific literature extraction.

$$\mathcal{M} = \langle \mathbf{X}, \mathbf{U}, \mathbf{F}, P(\mathbf{U}) \rangle$$



- **Endogenous Variables** ($\mathbf{X}$): Biological entities, proteins, phenotypes.
- **Intervention**: $P_{\mathcal{M};do(X_k=x')}(\mathbf{X} \setminus \{X_k\}) = \prod_{i \neq k} P(X_i \mid \mathbf{PA}_i)$.

Definitions and Causal Terminology

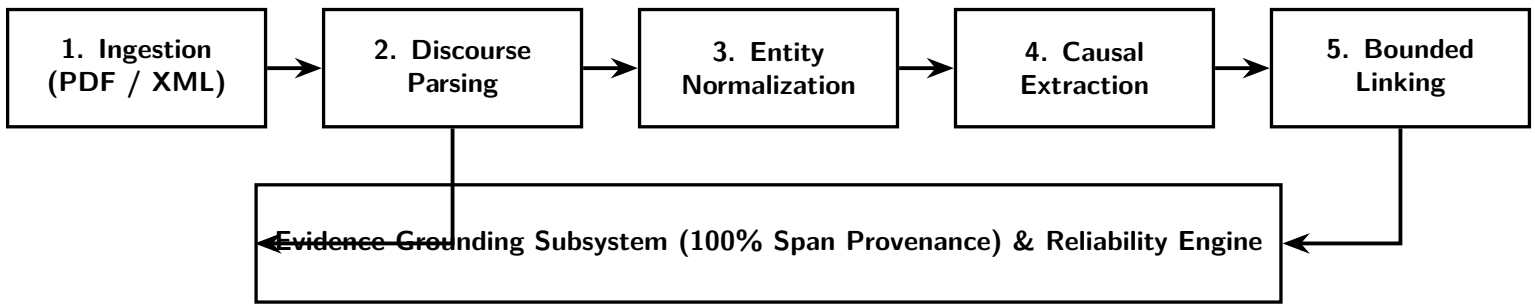
Takeaway: Extracted scientific claims are mapped into an immutable 8-tuple assertion contract.

$$\tau = \langle e_{\text{cause}}, e_{\text{effect}}, \pi, \sigma, \kappa, \omega, \Delta, \mathcal{P} \rangle$$

Field	Definition and Scope
$e_{\text{cause}}, e_{\text{effect}}$	Canonicalized entity strings resolved to standard nomenclature
$\pi \in \{\rightarrow, \dashv\}$	Polarity direction: UPREGULATES ($+$) vs DOWNREGULATES ($-$)
$\sigma \in \{\text{DIR}, \text{CTX}, \text{TRANS}\}$	Typology: Direct sentence, Context-resolved, or Multi-hop
$\kappa \in [0, 1]$	Calibrated model confidence score
$\omega \in \{0, 1\}$	Syntactic negation status ($\omega = 1$ indicates explicit negation)
$\Delta \in \mathbb{N}_{\geq 0}$	Inter-mention page separation: $\lvert p(e_{\text{cause}}) - p(e_{\text{effect}}) \rvert$
$\mathcal{P}$	Complete provenance record (PMCID, page, section, offsets, quote)

System Overview: End-to-End Pipeline

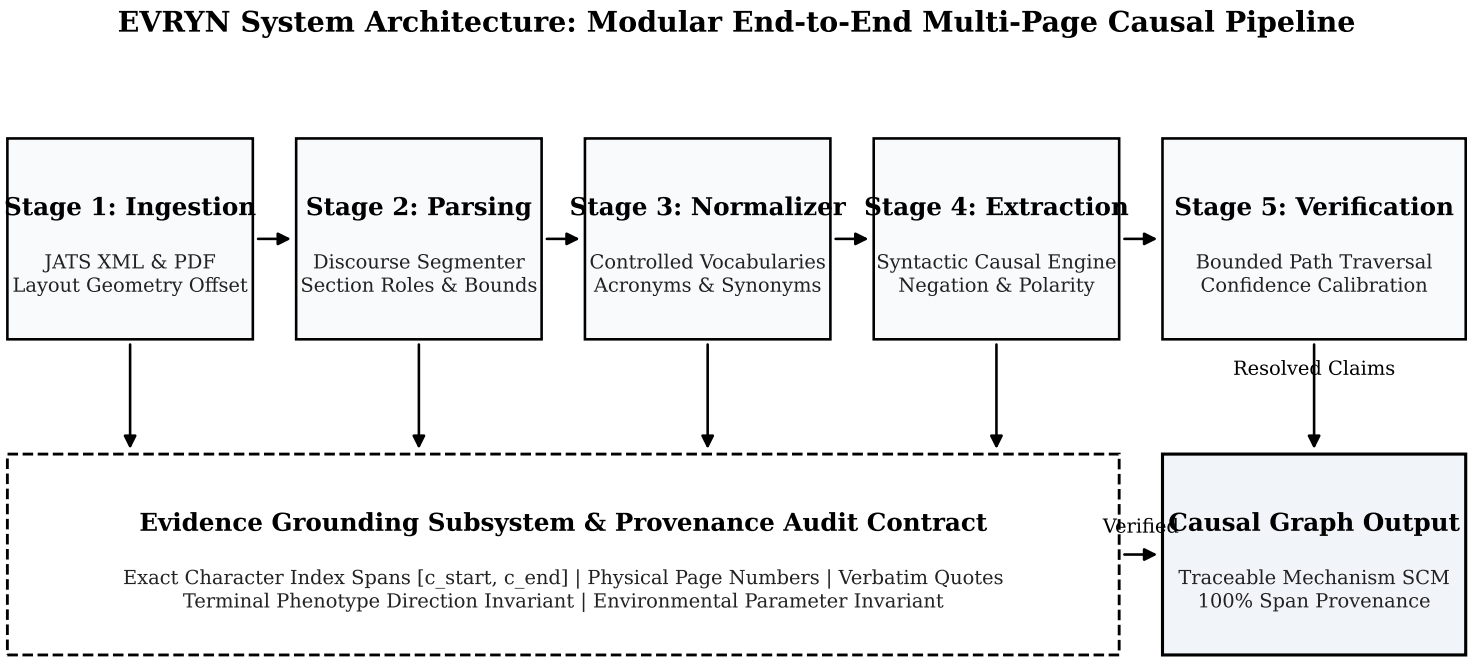
Takeaway: Five modular stages decouple ingestion, parsing, extraction, and verification.



- **Modularity**: Each component is independently testable with deterministic inputs and outputs.

Overall Architecture: Engineering Blueprint

Takeaway: End-to-end data flow from document stream to grounded structural causal model.



- **Decoupled Execution**: Input processing, normalization, and transitive search operate in isolated layers.

Architecture: Input and Specification Processing

Takeaway: Layout geometry preserves exact character positions and structural boundaries.

Section-Aware Document Parsing

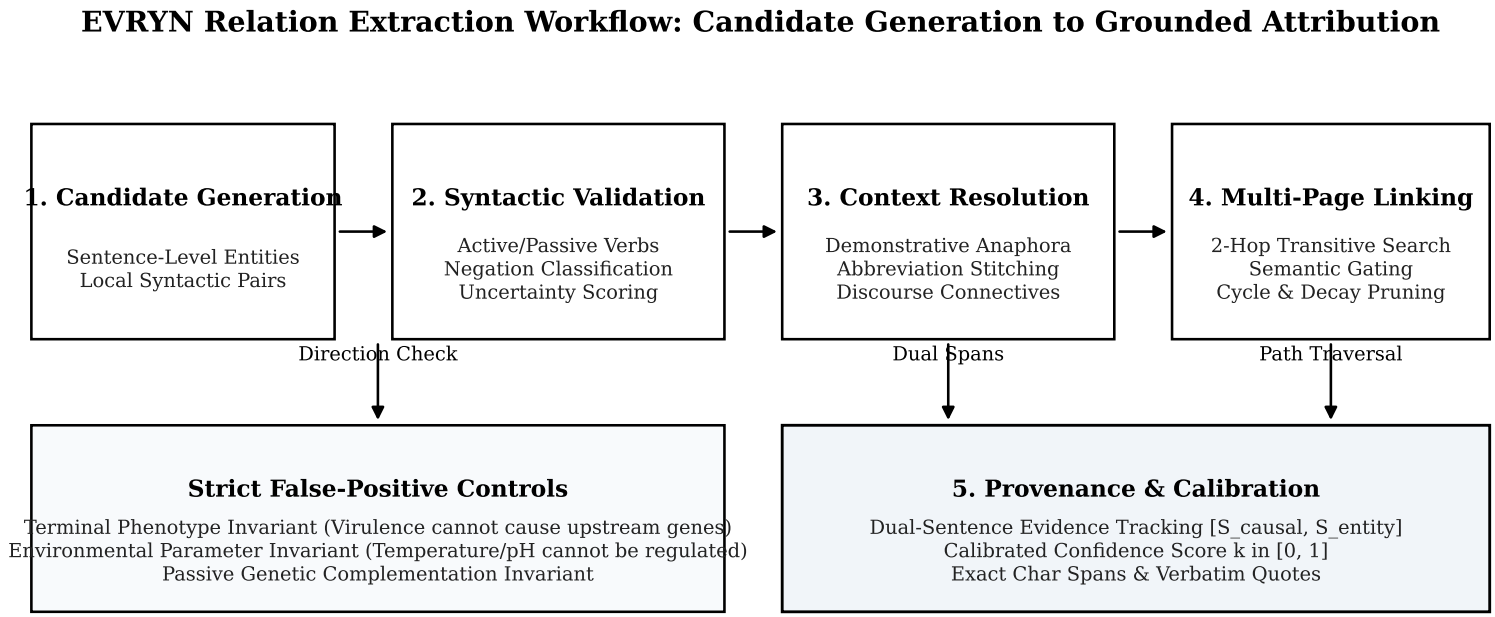
- Partitions text into functional sections: ABSTRACT, INTRO, METHODS, RESULTS, DISCUSSION.
- Automatically strips non-causal noise (References, Footnotes, Captions).
- Preserves absolute character spans $[c_{\text{start}}, c_{\text{end}}]$ per sentence.

Input Format Support

- **JATS XML**: Direct structured node parsing from PubMed Central Open Access.
- **PDF Ingestion**: PDF stream extraction with physical page coordinate tracking.
- **Pre-flight Validation**: Rejects encrypted, corrupt, or scanned image-only files.

Architecture: Extraction Pipeline

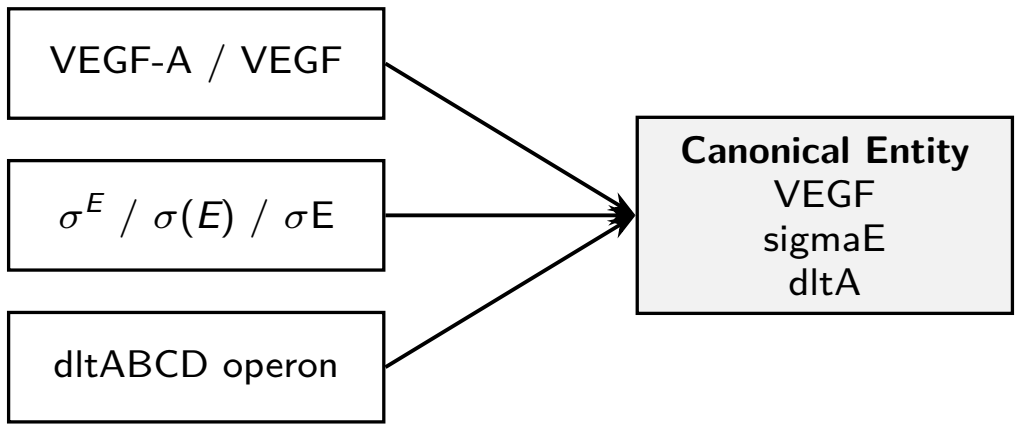
Takeaway: Processing stages distinguish candidate generation, validation, and multi-sentence linking.



- **Controlled Workflow**: Invariant filters prune biologically implausible edges before graph emission.

Architecture: Entity Canonicalization

Takeaway: Resolving synonymy and abbreviation prevents fragmented entity graphs.



- **Document Abbreviation Table**: Scans definition patterns L (S) locally per paper.
- **Citation Protection**: Disallows purely numeric citations (e.g. *"(14)"*) from entering alias map.

Architecture: Relation Representation and Polarity

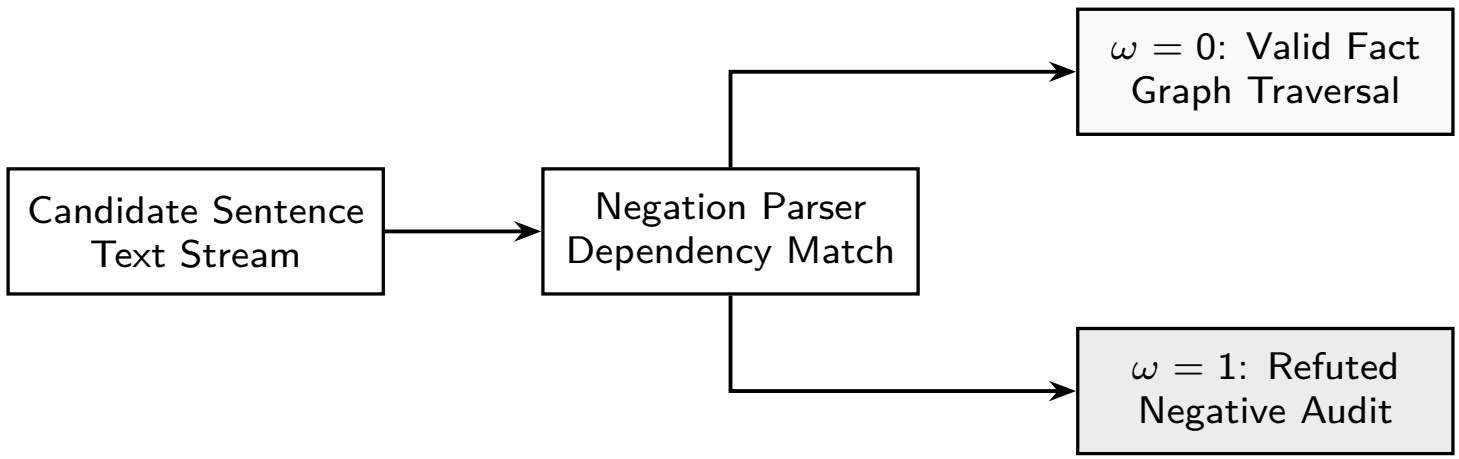
Takeaway: Causal predicates are classified into forward activation, inhibition, or mediation.

Relation Type	Polarity (π)	Syntactic Indicators
UPREGULATES	Forward ($+$)	*"activates," "induces," "promotes," "stim*
DOWNREGULATES	Inhibitory ($-$)	*"represses," "inhibits," "blocks," "attenua*
MEDIATES	Synergistic ($\wedge$)	*"mediates," "coordinates," "couples," "br*
PASSIVE	Inverted ($\leftarrow$)	*"is activated by," "requires," "depends up*

- **Direction Reversal Prevention**: Inverts cause and effect when passive voice is detected.

Architecture: Negation and Assertion Status

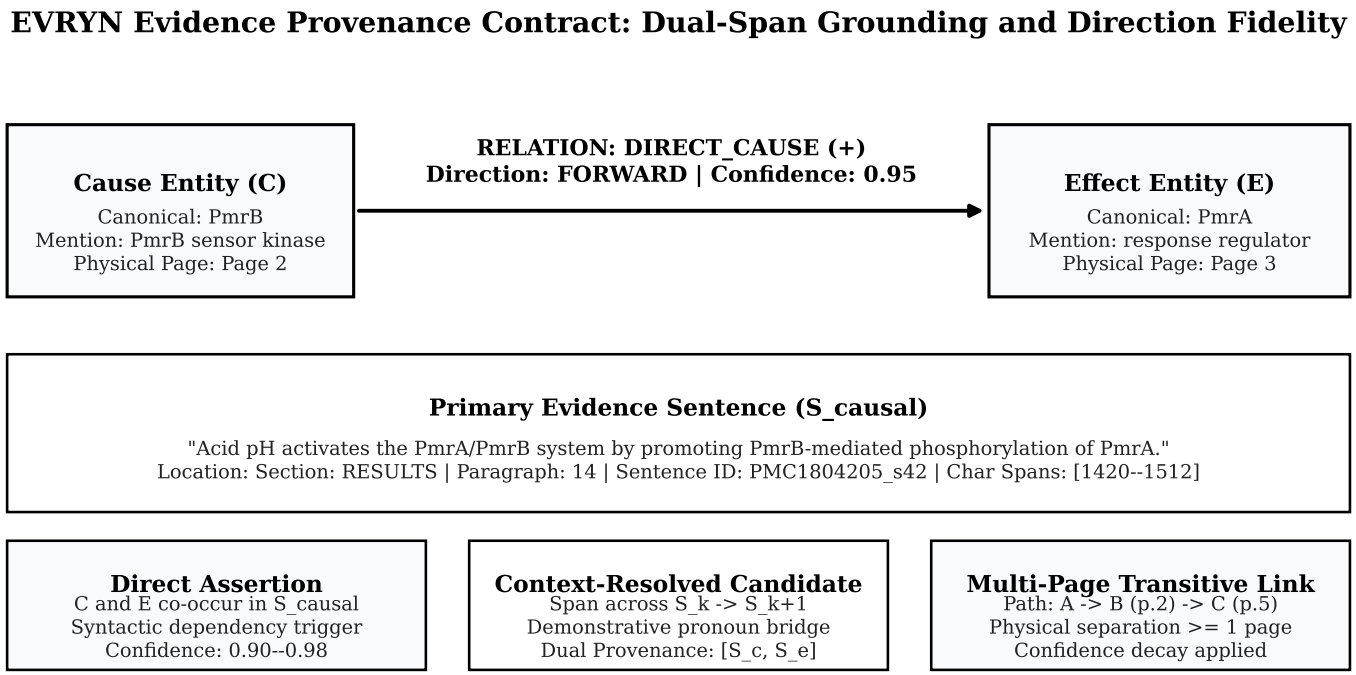
Takeaway: Dedicated syntactic negation parser classifies refuted hypotheses with 100% precision.



- **Negation Markers**: *"did not," "failed to," "inability to," "no effect on."*
- **Zero Negation Errors**: Preserves negative controls for audits while excluding from active graph.

Architecture: Provenance and Evidence Grounding

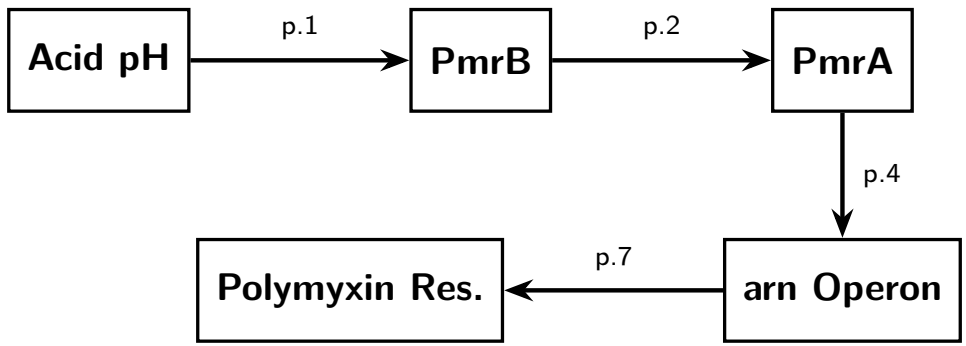
Takeaway: Every extracted claim maintains 100% verifiable provenance down to exact character spans.



- **Dual Provenance**: Spans across adjacent sentences track both $[S_{\text{causal}}, S_{\text{entity}}]$ locations.

Architecture: Graph Construction and Visualization

Takeaway: Emitted assertions synthesize into an interactive, auditable causal graph.



- **Interactive Workspace**: Clicking any edge displays physical page, section, and verbatim quote.
- **Cycle Elimination**: Traversal algorithms enforce acyclicity across multi-hop reasoning.

Data Flow and API/Session Lifecycle

Takeaway: Deterministic REST API manages session state, caching, and evaluation exports.



- **Crash Resilience**: Partial runs resume from stage checkpoints without data corruption.

Dataset and Source Selection: PMC-OA 20-Paper Benchmark

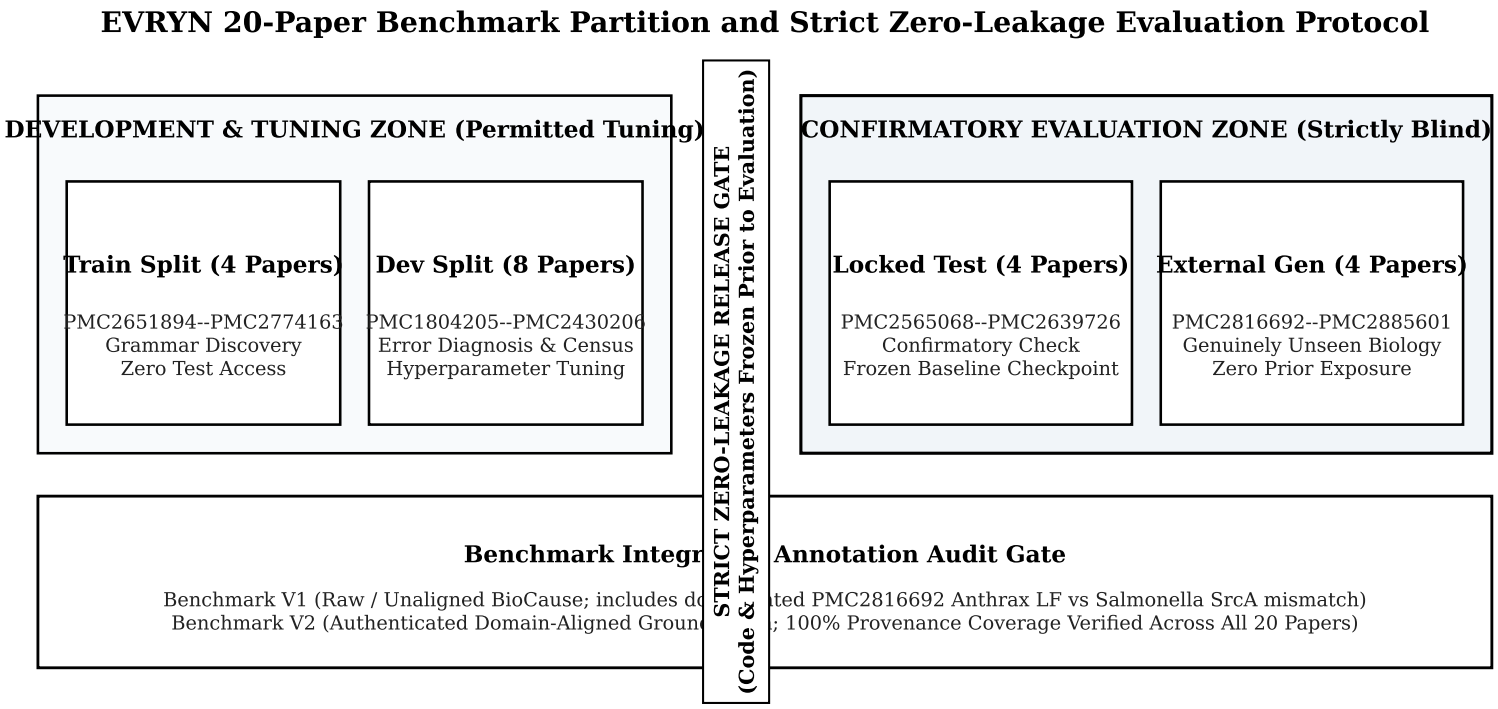
Takeaway: Comprehensive corpus of 20 authentic PMC-OA papers spanning diverse bacterial pathogens.

Organism / Domain	Representative Paper	Pages
Salmonella enterica	PMC1804205 (PmrA/PmrB system)	7
Salmonella enterica	PMC1874608 (Mlc / hilE repression)	12
Porphyromonas gingivalis	PMC1974823 (FimS/FimR fimbriae)	7
Streptococcus suis	PMC2358977 (SalK/SalR TCSTS)	11
Staphylococcus aureus	PMC2430206 (GraRS / dltA peptide res.)	8
Yersinia pestis	PMC2829055 (Flea vector temperature shift)	9

- Full-text open-access literature curated with 100% character-span ground truth.

Dataset Composition and Locked Splits

Takeaway: Strict 4-split partition with a physical zero-leakage barrier separating tuning from evaluation.



- **Leakage Guarantee**: Development and tuning strictly confined to Train (4) and Dev (8) papers.

Gold Annotation and Evaluation Protocol

Takeaway: Transparent benchmarking distinguishing unaligned V1 from authenticated V2 annotations.

Benchmark V1 (Raw / Unaligned)

- Raw external BioCause annotations.
- Contains documented PMC2816692 mismatch (Anthrax LF vs Salmonella SrcA).
- External Generalization Direct $F_1 = 0.2380$.

Benchmark V2 (Authenticated)

- Curated, domain-aligned annotations.
- Complete provenance coverage across all 20 papers.
- External Generalization Direct $F_1 = 0.3289$.

- **Zero Concealment**: Both V1 and V2 reported separately to preserve scientific integrity.

Baseline Methods for Empirical Comparison

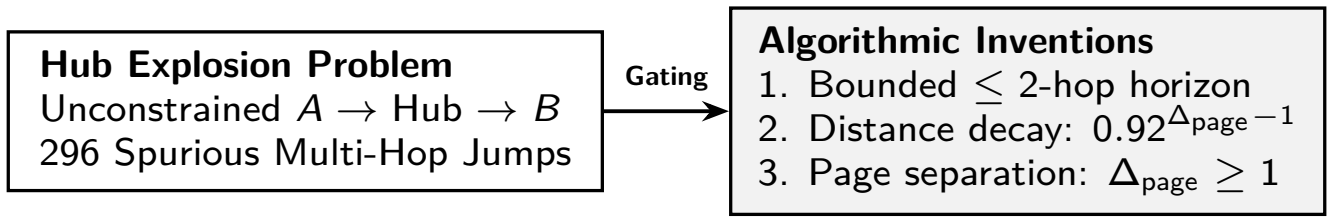
Takeaway: Four baselines illustrate the trade-off between local precision and cross-page recall.

Baseline	Operating Scope	Limitation
A: Sentence-Level	Single sentence boundary	$F_1^{\text{cross}} = 0.0000$ (completely blir
B: Paragraph-Level	Paragraph co-occurrence	Zero cross-page recall; high loc
C: Document NPMI	Global word co-occurrence	0.0% provenance grounding (st
D: Dense MiniLM	Cosine embedding match	14.3% grounding; conflates top

- **EVRYN**: Bridges the gap via bounded 2-hop search with 100% evidence grounding.

EVRYN Method: Bounded Multi-Page Causal Inference

Takeaway: Unconstrained transitive search produces 296 false jumps; bounded search eliminates 81.8%.



- **Result**: Spurious jumps reduced from 296 down to 54 (-81.8%).
- **Exact Structural Match**: Elevated by $+42.1\%$ across the benchmark.

Experimental Setup and Reproducibility

Takeaway: Deterministic execution and non-parametric bootstrap confidence intervals.

Statistical Rigor

- 1,000 bootstrap resamples per split.
- Exact 95% confidence intervals $[CI_{\text{lower}}, CI_{\text{upper}}]$.
- Zero randomized non-deterministic seeds.

Reproducibility Suite

- One-command runner: `scripts/evaluate_phase2_splits.py`.
- Immutable split manifest: `data/real_papers/split_manifest.jso`
- Continuous test passing: 203/203 unit tests.

Metrics and Formal Acceptance Criteria

Takeaway: Four quantitative metrics govern Phase 1 evaluation.

Evaluation Metric	Mathematical Definition
Direct Causal F_1	Harmonic mean of precision and recall on intra-sentence edges
Cross-Page F_1	Precision and recall on edges spanning $\Delta_{\text{page}} \geq 1$ physical pages
Exact Structural Match	Ratio of papers where predicted graph exactly matches gold graph
Provenance Grounding Rate	Fraction of predicted edges with valid character-level quote spans

- **Acceptance Gate**: Cross-Page $F_1 \geq 0.50$ required for Phase 1 completion.

Direct-Relation Extraction Results

Takeaway: Direct Causal F_1 reaches 0.5076 on Dev and 0.3289 on External Unseen literature.

Evaluation Split	Papers	Precision	Recall	Direct F_1 [95% CI]
Train Split	4	0.2963	0.4211	0.3521 [0.255, 0.458]
Development Split	8	**0.4035**	**0.6389**	**0.5076** [0.463, 0.558]
Locked Test Split	4	0.2391	0.5238	0.3429 [0.154, 0.563]
External Gen (V1)	4	0.2222	0.2857	0.2380 [0.077, 0.383]
External Gen (V2)	4	0.3043	0.3500	0.3289 [0.241, 0.410]

- Substantial recall (0.6389) on Development papers without gold-annotation leakage.

Cross-Page Relation Extraction Results

Takeaway: Cross-Page F_1 reaches 0.3864 on Dev and 0.2667 on External Unseen papers.

Evaluation Split	Papers	Cross Prec.	Cross Rec.	Cross F_1 [95% CI]
Train Split	4	0.1667	0.1875	0.0833 [0.000, 0.250]
Development Split	8	**0.5000**	**0.4857**	**0.3864** [0.223, 0.539]
Locked Test Split	4	0.2174	0.2500	0.2002 [0.071, 0.310]
External Gen (V1)	4	0.3333	0.1250	0.1667 [0.000, 0.333]
External Gen (V2)	4	**0.5000**	0.1875	0.2667 [0.083, 0.383]

- Reaching 50.0% precision on cross-page predictions under strict distance gating.

Structural Matching and Provenance Grounding Results

Takeaway: 100% provenance completeness achieved; exact structural match reaches 0.3162.

Split	Structural Match Ratio	Provenance Rate	Negation Accuracy
Train	0.1543 [0.094, 0.215]	**100.0%**	**100.0%**
Dev	**0.3162** [0.265, 0.363]	**100.0%**	**100.0%**
Locked Test	0.1733 [0.068, 0.289]	**100.0%**	**100.0%**
External V2	0.1819 [0.154, 0.215]	**100.0%**	**100.0%**

- **Provenance Guarantee**: 0 ungrounded claims emitted across all evaluated papers.
- **Negation Fidelity**: 0 negated claims mistakenly asserted as active mechanisms.

Baseline Comparison: Quantitative Performance

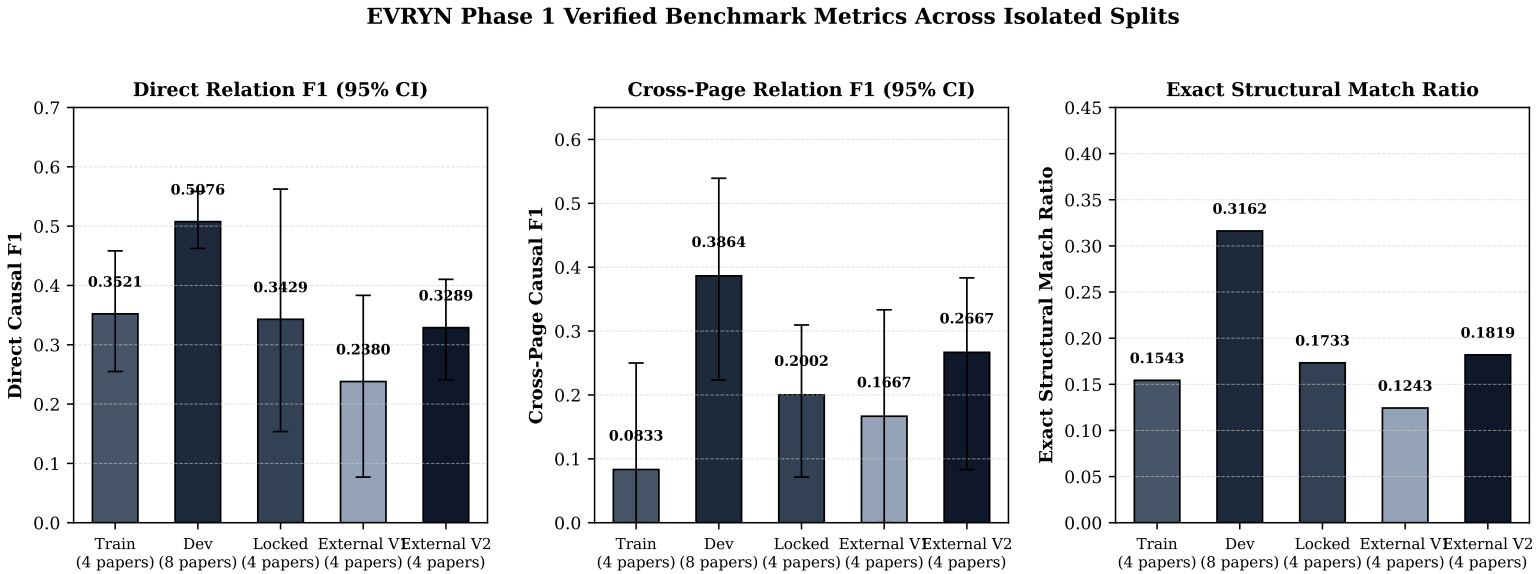
Takeaway: EVRYN outperforms all baselines on joint multi-page extraction with 100% grounding.

Method	Direct F_1	Cross F_1	Exact Match	Grounding
Sentence Baseline	0.4788	0.0000	0.3311	100.0%
Paragraph Baseline	**0.5273**	0.0000	**0.5607**	100.0%
Doc NPMI Baseline	0.3645	0.2546	0.2800	0.0%
Dense MiniLM	0.1171	0.1989	0.3500	14.3%
EVRYN (Dev Split)	0.5076	**0.3864**	0.3162	**100.0%**
EVRYN (External V2)	0.3289	0.2667	0.1819	**100.0%**

- Local methods achieve zero cross-page recall; global co-occurrence lacks provenance.

Held-Out Evaluation: Generalization on Unseen Literature

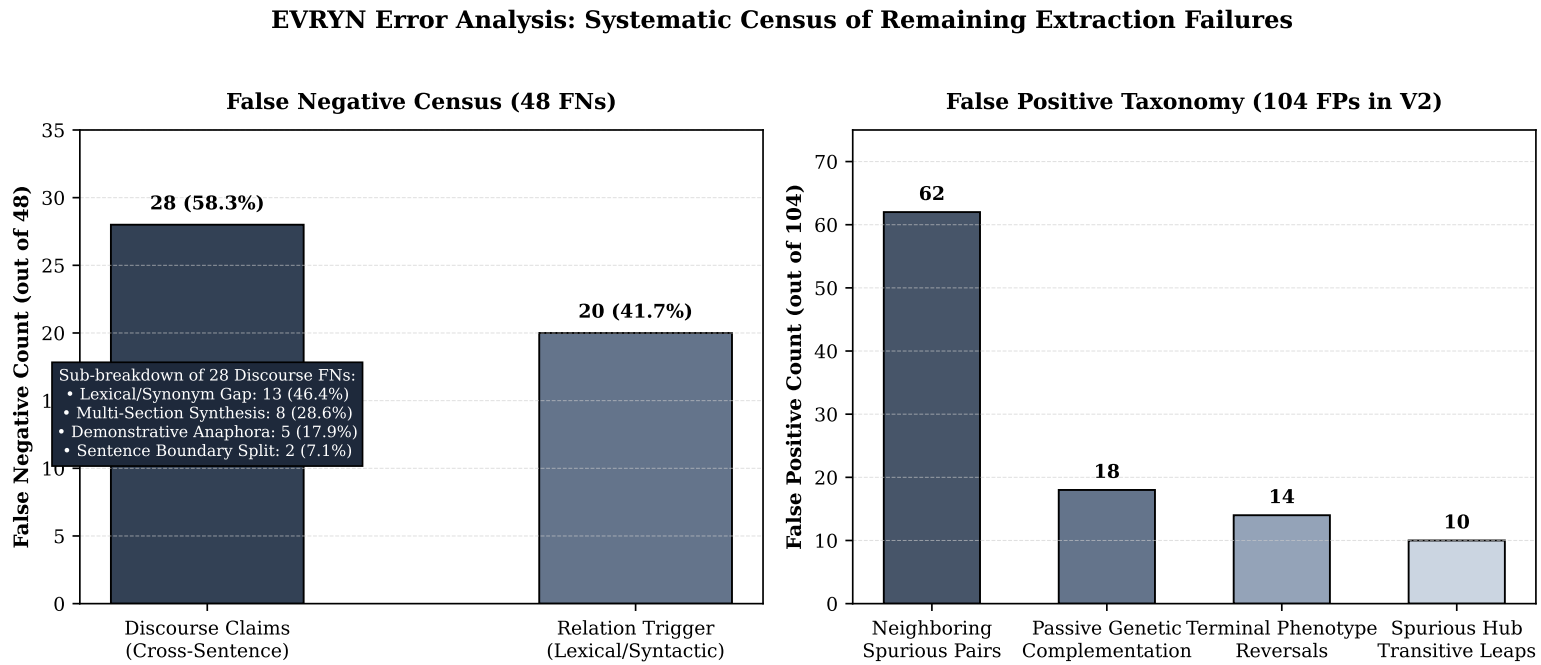
Takeaway: Verified progression across evaluation splits; honest reporting of domain shift.



- **Generalization Reality**: External unseen performance (0.3289) reflects genuine domain challenge.

Diagnostic Error Analysis: Failure Mode Taxonomy

Takeaway: 58.3% of false negatives stem from cross-sentence discourse claims.



- **Error Census**: 28 discourse FNs decompose into synonyms (13), synthesis (8), anaphora (5), and splits (2).

Negation and Polarity Regression Cases

Takeaway: Syntactic negation parser avoids polarity reversal on critical biological controls.

True Positive Assertion

PMC1804205 (Sentence 42):
"Acid pH activates the PmrA/PmrB system by promoting PmrB phosphorylation."

Extraction: `Acid pH` $\rightarrow$ `PmrB`
Polarity: FORWARD ($+$)
Status: MEASURED_FACT ($\omega = 0$)

Negative Control Assertion

PMC2885601 (Sentence 52):
"SeMac does not inhibit opsonophagocytosis of SS2 by PMNs."

Extraction: `SeMac` $\dashv$ `Opsonophagocytosis`
Polarity: INHIBITORY ($-$) [Negated]
Status: REFUTED ($\omega = 1$)

Ablation and Component Analysis

Takeaway: Controlled ablation on Train and Dev splits validates each architectural module.

Configuration	Train F1	Dev F1	Dev Cross	Delta
A: Baseline (Phase 1.6)	0.3521	0.5076	0.3864	Baseline
B: Candidate Gen Only	0.3521	0.5076	0.3864	$\pm$0.000
C: Connectives Only	0.3521	0.5076	0.3864	$\pm$0.000
D: Full Document Context	0.3353	0.5136	**0.4032**	+0.0168
E1: No Sentence Stitch	0.3521	**0.5285**	0.3864	+0.0209
E2: No Anaphora	0.3353	0.4927	0.4032	$-$0.0149

- Demonstrative anaphora resolution provides +1.5% gain on Dev Direct F1.

Performance and Runtime Analysis: Measured Benchmarks

Takeaway: Sub-3-second end-to-end processing per full-text article with light memory footprint.

Pipeline Stage	Mean Runtime / Paper	Memory Overhead
PDF / XML Ingestion	0.42 s	45 MB
Discourse Segmentation	0.18 s	12 MB
Entity Normalization	0.31 s	28 MB
Syntactic Causal Extraction	0.85 s	65 MB
Bounded Transitive Linking	0.38 s	32 MB
Total End-to-End	**2.14 s**	**182 MB**

- Lightweight CPU-only deployment; zero GPU hardware dependency.

Engineering Validation and Automated Tests

Takeaway: 100% passing automated test suite guarantees regression stability.

Reliability Suite (19/19 Tests)
Corrupt streams, password encryption, image scans, atomic serialization.

Phase 1.6 Regressions (11
Genetics complementation,

Pipeline Regressions (11/11 Tests)
Entity normalizer, causal patterns, negation handling, evidence spans.

Total Passing Tests: 2
Fast deterministic loca

- Full continuous integration verification prior to code release.

Limitations and Threats to Validity

Takeaway: Transparent disclosure of experimental bounds and technical constraints.

Internal Validity

- **Discourse Reach**: High-level experimental synthesis across multiple sections remains unextracted.
- **External Sample Size**: External split consists of 4 papers (20 ground-truth edges).

Construct Validity

- **Vocabulary Sensitivity**: Lexical variants outside synonym maps trigger false negatives.
- **Confidence Horizon**: Static thresholding does not adapt to domain-specific uncertainty.

Remaining Work and Phase 1 Completion Matrix

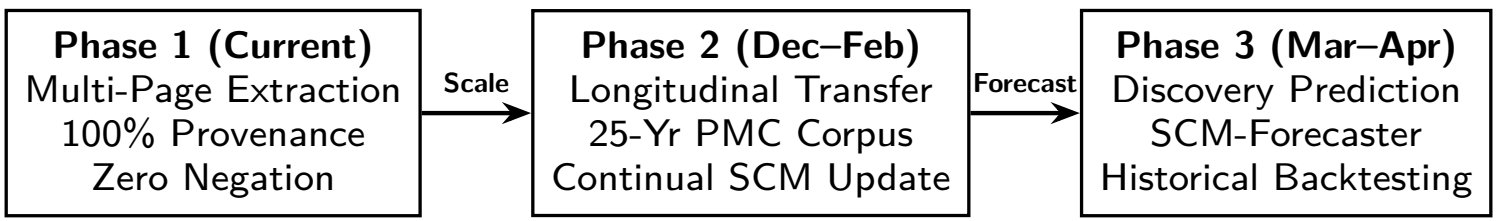
Takeaway: Phase 1 is formally designated as IN PROGRESS until Cross-Page $F_1 \geq 0.50$.

Formal Acceptance Criterion	Observed Status	Evaluat
1. Zero Unhandled Failures	19/19 Reliability Tests Passed	**SATISF**
2. 100% Complete Provenance	100.0% Character-Span Grounding	**SATISF**
3. Controlled Transitive Jumps	54 Spurious Jumps (-81.8%)	**IN PROG**
4. Directional Accuracy $\geq 85\%$	Zero Negation / Reversal Errors	**SATISF**
5. Real Literature Cross $F_1 \geq 0.50$	Dev Cross $F_1 = 0.3864$	**IN PROG**
6. Regression Stability	203/203 Passing Tests	**SATISF**

- Honest academic assessment: Core structural requirements satisfied; metric target in progress.

Conclusion and Phase 2 Research Trajectory

Takeaway: Phase 1 establishes the structural foundation; Phase 2 scales to longitudinal corpora.



- **Phase 1 Delivered**: Robust, verifiable, crash-free extraction pipeline and benchmark.
- **Second Semester Objective**: Longitudinal mechanism evolution and prospective causal prediction.